Mill Spindle - Troubleshooting Guide

Recently Updated

LAST UPDATED: 10/14/2025

TG0101

Revision B - 11/2025

Introduction

Download and fill out the Spindle Inspection Report Checklist below before replacing any parts.

Symptom Table

Symptom	Possible Cause	Corrective Action
Spindle not turning and at Power On: A Warning appears " The spindle may be rotating freely without power... "	The software version is outdated.	Check the software version to be 100.20.000.1200 or higher. Contact your Haas Factory Outlet to have the software updated.
Noise during rigid tapping.	A NGC software error in version 100.17.000.xxx that caused a noticeable growling noise during rigid tapping, particularly with coarse pitch threads.	Update software to the latest version.
	Problems with the spindle encoder.	Look for runout, loose mounting screws or shaft clamps, and any other obvious problems. Spindle Non-Contact Encoder (NCE) - Troubleshooting Guide - NGC

The Spindle is seized or it won't turn.	The spindle has been crashed and now damage.	Review alarm history for indications of a crash, check for visual damage to the spindle
	There is heavy side loads causing the cap to gall the shaft.	Check for clearance all the way around with a piece of 0.001” shimstock; loosen the cap, re-center and tighten.
	The spindle lubrication system is not functioning correctly.	Inspect the spindle lubrication system.
	Spindle installed incorrectly	Check Motor coupler for damage or indentations at the pin interfaces. Contact local HFO about issue including relevant pictures of the spindle to motor coupler.
The spindle temperature exceed 140° F (60° C)	The spindle lubrication system does not function correctly.	Check for a clog in the spindle lube pump and verify the type of oil.
The spindle makes an unusual noise.	The spindle drive belt is worn or damaged.	Check the spindle drive belt for damaged.
	The encoder pulleys or drive belt are worn or damaged.	Check the encoder pulleys and belt.
	The spindle lubrication system does not function correctly.	Check for a clog in the spindle lube pump and verify the type of oil.

The surface finish shows chatter, or the spindle vibrates. **Alarm 966** EXCESSIVE PART/TOOL IMBALANCE.	The Main Processor PCB standoff hardware is loose or the Control Cabinet mounting hardware is loose or causing a bind.	Tighten the Main Processor PCB standoff hardware. Loosen the Control Cabinet mounting hardware, snug the hardware, then torque down the hardware.
	The tool or tool holder may be too long for the application.	Longer tools are less stable than shorter tools. Always use the shortest tool possible, and remember that a 10% reduction in the length-to-diameter ratio results in a 25% increase in tool stiffness. Unless absolutely necessary, the tool's stick-out from the holder should not be more than 3 times longer than its diameter.
	The spindle drive belt is worn or damaged.	Check the spindle drive belt.
	The encoder pulleys or drive belt are worn or damaged.	Check the encoder pulleys and belt.
	The toolholder is fretting.	Verify the length of the tool with the manufacturer's specifications.
	The drawbar force is incorrect.	Make sure the drawbar clamp force is correct. Refer to: VMC - Spindle - Drawbar - Force Measurement

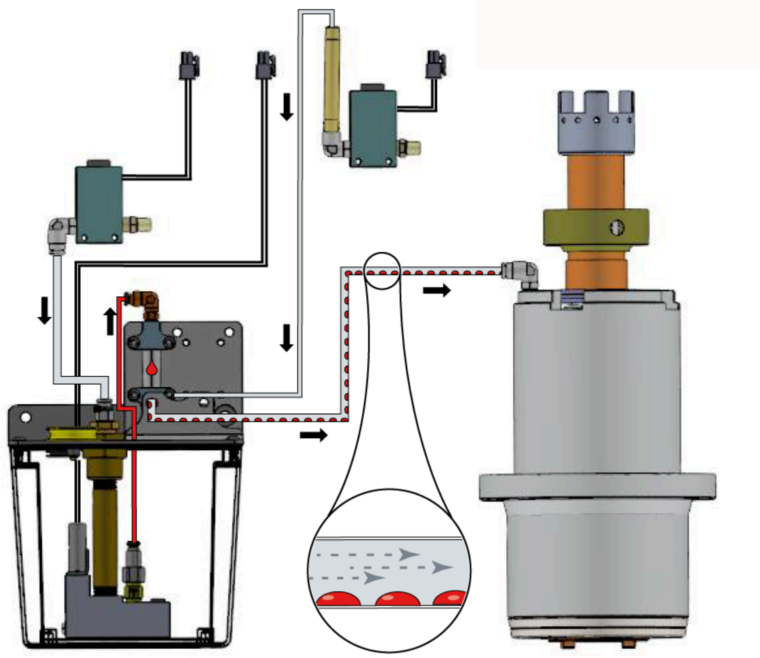
	The fixture does not support the part.	Adjust the cutting load or program.
	The tool is unbalanced.	Balance the tool holder with the cutting tool in it. If you cannot balance the tool assembly, replace it with one that you can balance.
	Excessive Endmill Runout - Mill.	Tool runout should not exceed 0.0003" (0.076 mm). To check the tool runout, place a dial indicator on the tool, and then rotate the spindle by hand.
	The spindle bearings are damaged.	Perform a vibration analyzer.
	The spindle is not balanced.	Contact your HFO to balance the spindle.

Tools stick or do not come out of the spindle easily.	The tool stayed in the spindle for hours while the machine was not in operation.	Make sure a tool is not in the spindle when the machine is not in operation.
	The volume of air supplied to the machine is not sufficient.	Refer to: New Machine / Pre-Installation information for the correct air requirements.
	The drawbar force is incorrect.	Make sure the drawbar clamp force is correct.
	The toolholder is fretting. Damage tool shank or spindle taper.	Verify the length of the tool with the manufacturer's specifications. Maintain and inspect tool holders, and spindle taper.
	There is no grease on the pull stud.	Lubricate the pull stud.
	(Through-spindle coolant only) Some coolants cause tools to stick.	Coolants containing paraffin waxes can causing sticking tools when using high pressure Through Spindle Coolant with small orifice tooling. Test with a different type of coolant.
	The spindle is getting too hot.	Test the spindle lubrication system. Lack of, or excessive oil will create heat in the spindle.
	Long gage length tooling.	Stub up the tools to the extent possible. Reduce depth of cut. Use dynamic tool path and constant cutter engagement.

While running there is bearing Noise: squealing, screeching or skating.	Noise from imbalance and vibration	Make sure the tools are balanced.
	Squealing noises from the TSC union.	Check the TSC union.
Rubbing noises; Heat at the front cap.	Spindle cartridge has been bumped there is a chance the front cap has been pushed to one side and is rubbing on the nose of the spindle shaft.	Check for clearance all the way around with a piece of 0.001” shimstock; loosen the cap, re-center and tighten.
Noises during tool change.	Some models have had the precharge circuit eliminated.	This is a reliability improvement but a noise can be heard when the TRP contacts the drawbar or the release ring. This is normal.
The spindle oscillates during spindle orientation.	The Wye contactor (if equipped) coil is not receiving power.	Make sure the Wye/Delta contactors are operating correctly. Refer to: Wye-Delta Contactor - Troubleshooting Guide.
	The spindle motor wiring is incorrect.	The spindle motor wiring is incorrect. Refer to: Spindle Motor - Wiring Configurations.
	The vector drive is defective.	The vector drive is defective. Refer to: Vector Drive - Troubleshooting Guide.

Alarm 174 Tool Load Exceeded	Tool Load limit incorrect	View the Advanced Tool Managment (ATM) tab to adjust user defined tool load limit
	Aggressive feedrates	Adjust your program for a more conservative feedrate
Machine is experiencing surface finish issues	Preload set to the bearings may be incorrect	Perform the Spindle Deflection Test - Mill procedure for preload verification **NOTE:** Only perform this procedure AFTER all other troubleshooting steps outlined in the Mill Spindle Inspection Report, linked at the top of this page, have been performed.
Lube line slips out of fitting easily.	Plastic fitting has failed.	Replace fittings with updated steel version. Order **PN 58-2196.**

Lubrication



Corrective Action:

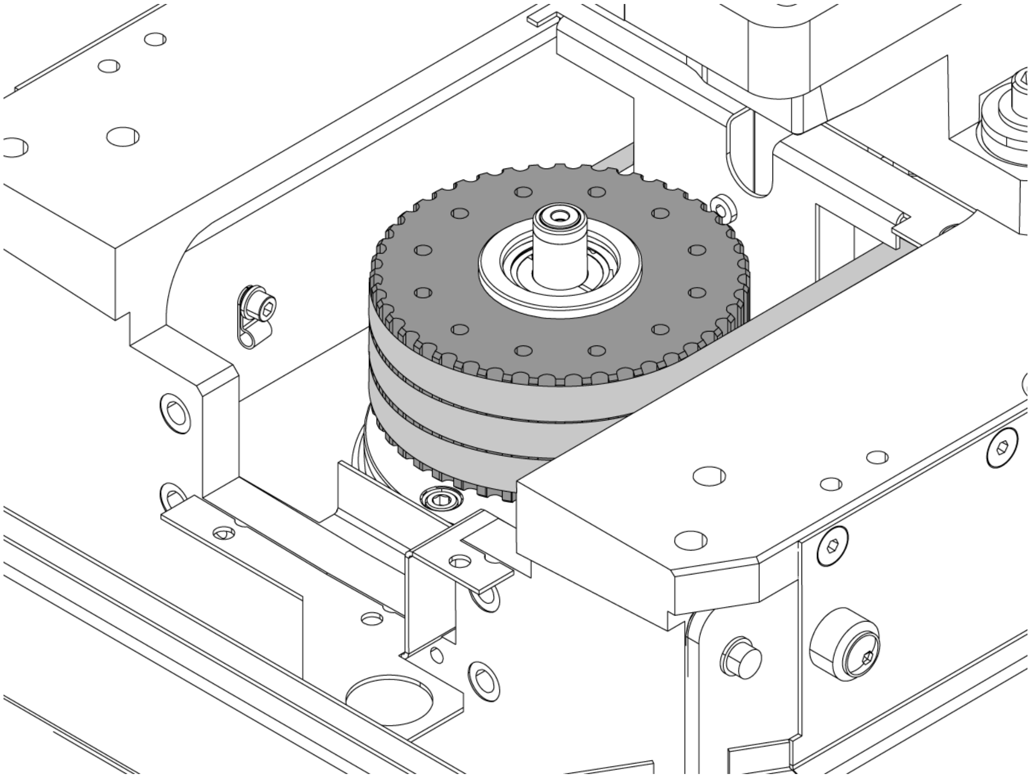
Insufficient or excessive oil supply to the spindle bearings can cause the spindle to overheat. Using the incorrect spindle lubrication oil on the spindle bearings can cause the spindle to overheat. Make sure you use the correct spindle lubrication (SHC 625) and the air pressure to the spindle is correct.

Inspect all tubing, hoses and fittings in the lubrication system for leaks. Check for puddles of oil to help locate leaks.

Go to HAAS SPINDLE MINIMUM LUBRICATION TROUBLESHOOTING for additional troubleshooting information

Go to MECHANICAL BIJUR LUBE PUMP - TROUBLESHOOTING GUIDE for additional troubleshooting information.

Drive Belt



Corrective Action:

Power off, lock and tag out machine. Remove the spindle head covers. Disconnect the air supply to the machine. Remove the four 3/8-16 SHCS on the TRP. Remove the TRP and set it aside.

Inspect the spindle drive belts and pulleys for damage: Look for frayed, stretched or broken belts, and damaged or stripped cogs on the inside surface of the belt. Refer to the DRIVE BELT - TROUBLESHOOTING GUIDE.

Encoder Pulley/Belt

Corrective Action:

Make sure the encoder belt is adjusted correctly and is not damaged. Replace a damaged or worn belt.

Make sure the encoder pulley is not worn. Make sure the set screw is tight.

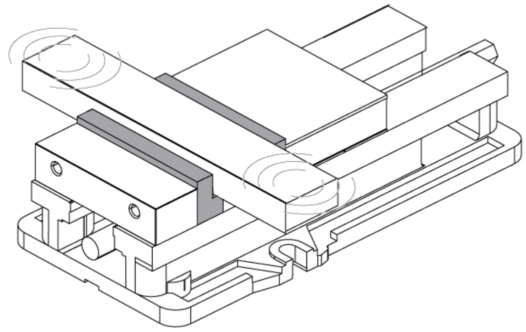
Reference SPINDLE NON-CONTACT ENCODER (NCE) - TROUBLESHOOTING GUIDE - NGC for more information about the encoder assembly.

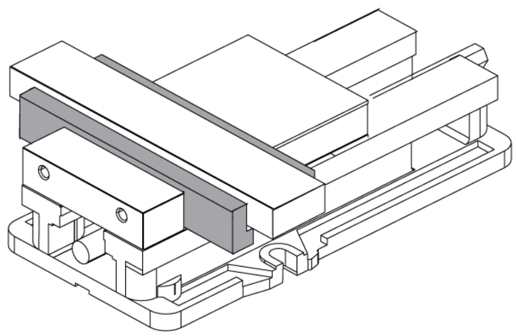
Coolant

Corrective Action:

If the machine has through-spindle coolant (TSC), different types of coolant can cause the toolholder to stick in the spindle. Consider a different type of coolant if the symptom persists.

Workholding





If the workholding does not adequately support the part, it can cause chatter.

Corrective Action:

Stiffen the workholding if possible.

Lower the cutting load with adjustments to the tool selection, or the program.

Balance

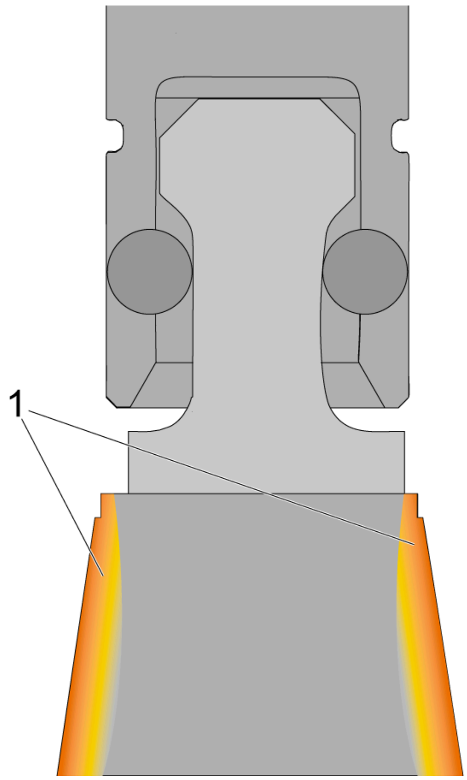
Corrective Action:

Make sure the spindle is balanced. Your local HFO can balance these spindles:

- All belt-driven spindles
- Inline spindles made after August, 2016

All tooling that runs over 10K RPM must be balanced. Do not exceed the manufacturer recommendations for special tooling.

Toolholder Is Sticking



This effect can occur while the machine is in operation if a cold toolholder is placed into a warm spindle taper. This creates thermal expansion [1] on the sides of the toolholder and makes it stick in the spindle taper before it is released.

A common example is if you change from a hot cutter to the cool spindle probe to check your part at the end of a cycle. If the cool probe is left in the hot spindle overnight, it will make a "popping" noise when it is released.

Corrective Action:

Clean the toolholders and spindle taper. Check for damaged toolholders before you put them back in the machine. A damaged toolholder can cause damage to the spindle taper and create tool runout or problems with finish.

Make sure the pull stud is lubricated. Refer to the MILL - TOOLING - MAINTENANCE procedure.

Toolholder Is Fretting

Corrective Action:

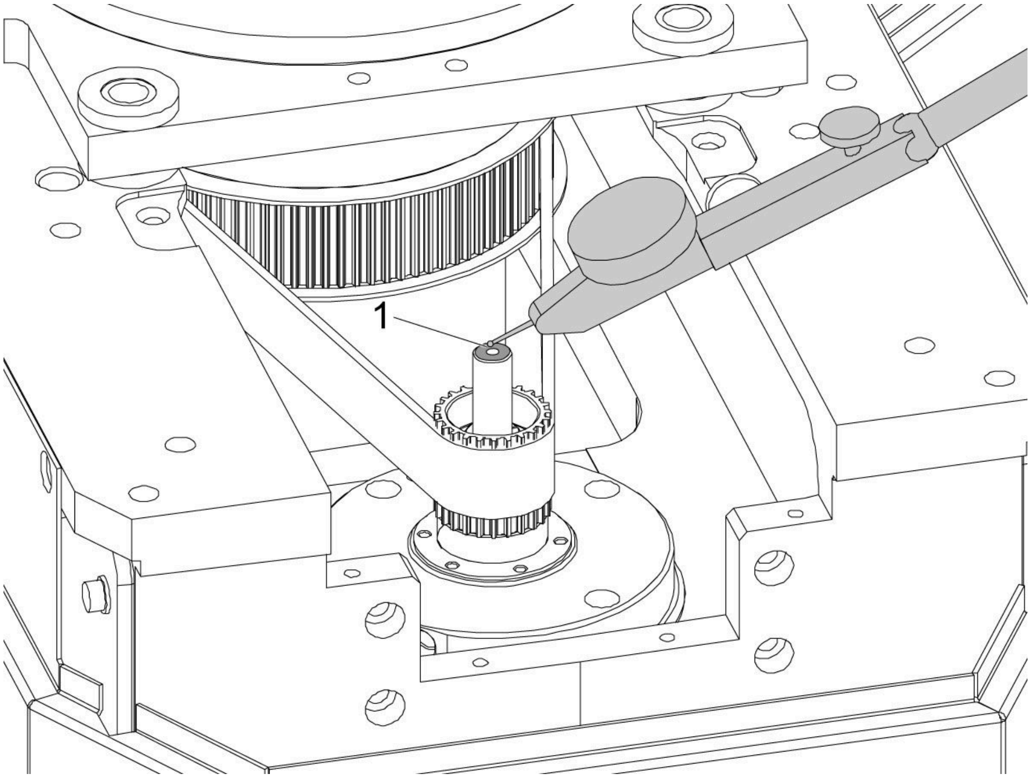
Look for fretting on the toolholders. Long tools can cause fretting on toolholders if you use them aggressively. If there is fretting on the toolholder, or if the symptom persists, go to SPINDLE - DRAWBAR - TROUBLESHOOTING GUIDE.

Bearings

Corrective Action:

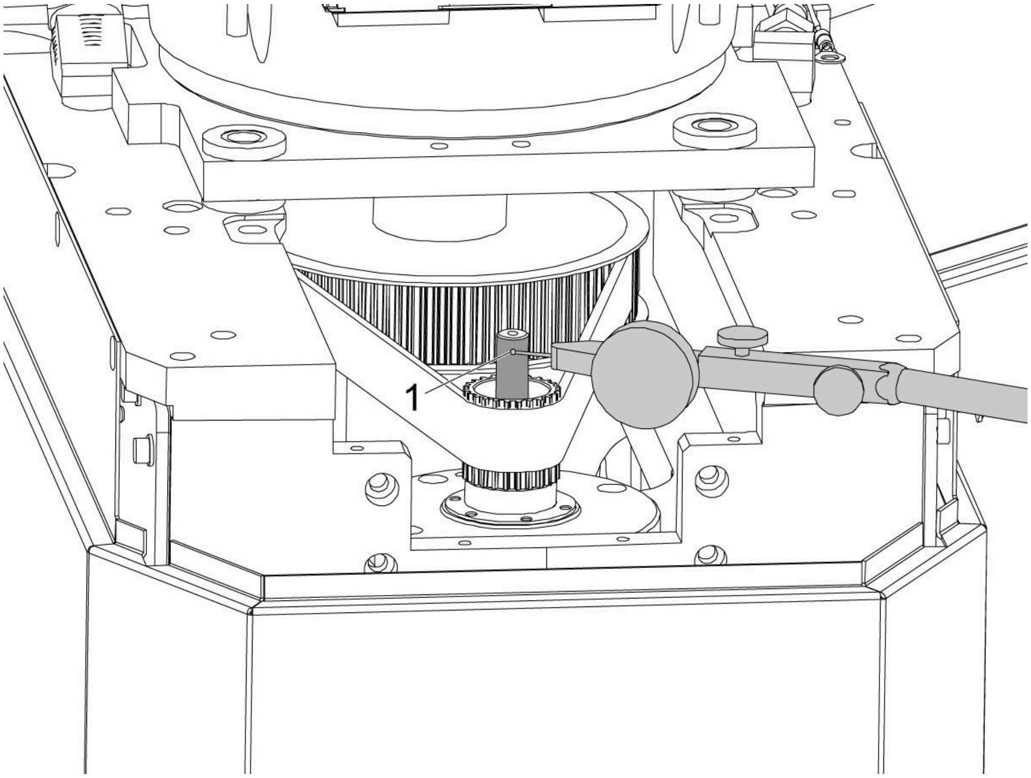
Damaged bearings in the spindle can cause finish issues on the part. Contact your local HFO for a vibration analysis to see if the bearings are damaged.

Axial Check



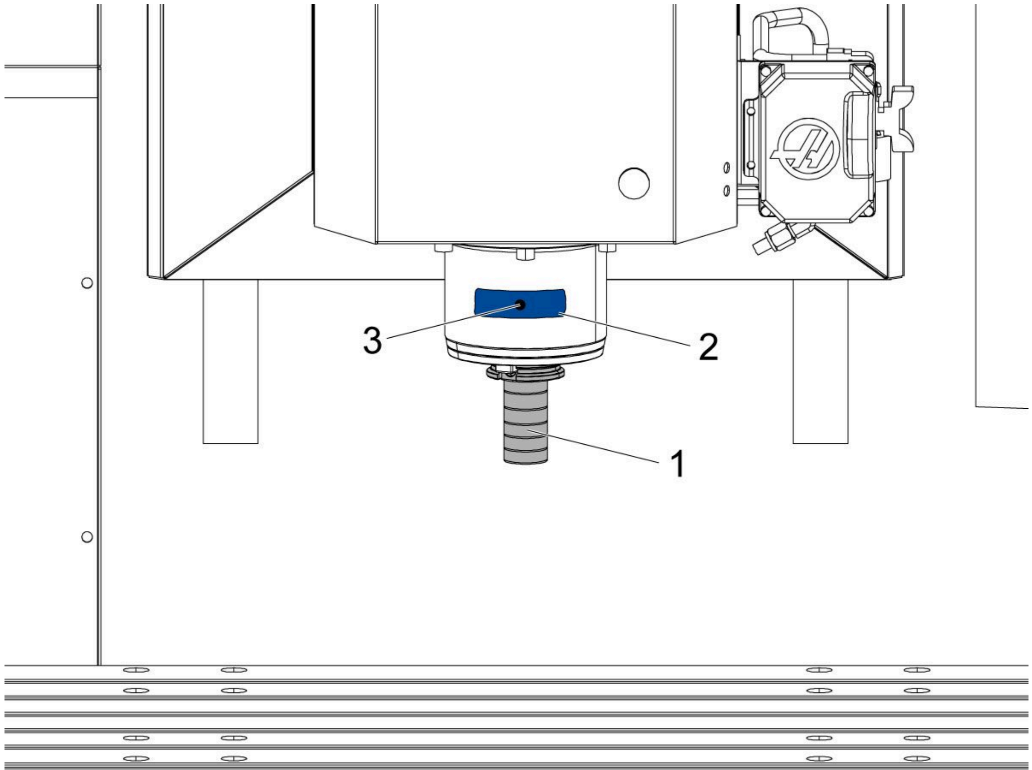
For belted spindles, place the indicator on the top face of the drawbar to check the axial runout.

Radial Check



For belted spindles, place the indicator on the shaft of the drawbar to check the radial runout.

Spindle Temperature Check

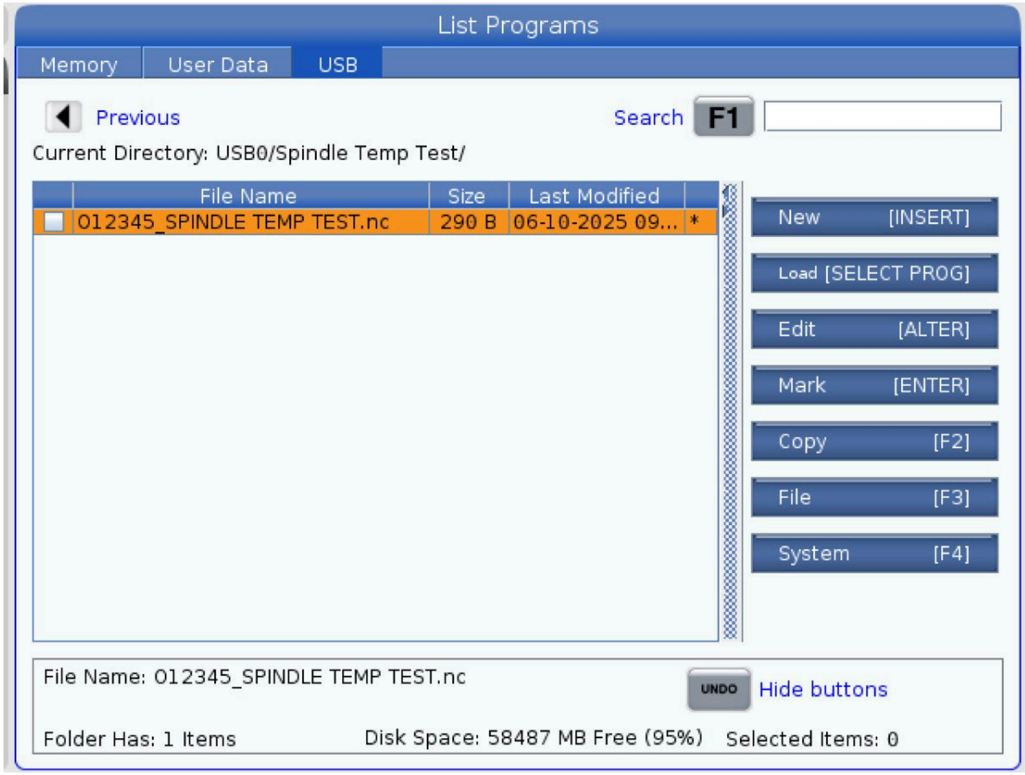


1

Note: DO NOT operate the spindle without a dummy tool in the taper [1]. Running the spindle without a tool will damage the bearings.

Note: Perform a spindle lubrication test and run spindle warm-up program before proceeding.

Place a piece of masking tape [2] on the front side of the spindle. Draw a black dot [3] in the center of the tape. Take the thermal readings from this location.



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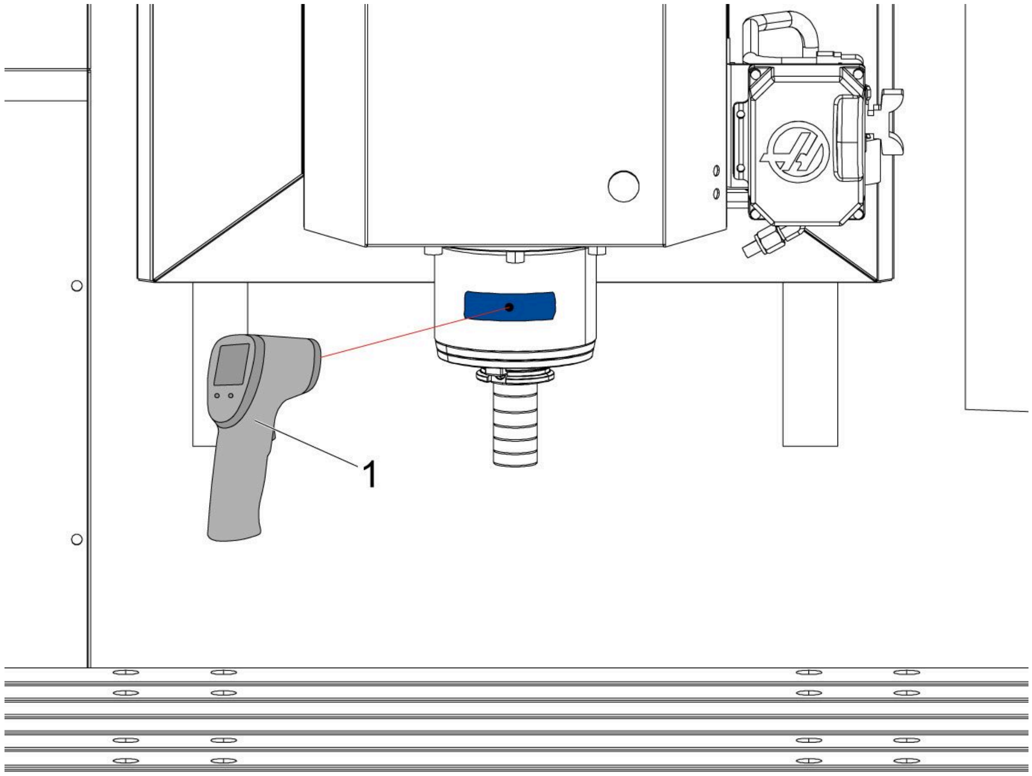
Download the program listed below onto a USB.

Spindle Temp Test:

O12345

Install the USB with the downloaded program into the USB port on the side of the pendant.

Load and run program **O12345.**



3

After the program is done running (40 minute run time), use a temperature gun [1] to measure the spindle temperature.

Verify the spindle temperture does not exceed 140° F (60° C).